UNIVERSITY OF TUSCIA

Department of Economics, Engineering, Society and Business Organization



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DEGLI STUDI DELLA
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MASTER'S DEGREE IN MECHANICAL ENGINEERING

VIRTUAL PROTOTYPING COURSE
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IDEATION, VALIDATION AND PROTOTYPING OF AN ESSENTIAL OIL EXTRACTOR

Introduction

This report documents the conception, virtual prototyping, and preliminary analysis of a kitchen-scale essential oil extractor developed within the Virtual Prototyping course. The project was structured to emulate a simplified industrial product development process, integrating engineering design activities with basic techno-economic and environmental considerations.

First, a simplified market investigation was conducted to justify the need for a domestic extractor, identifying potential users, typical operating scenarios (small batches, limited available space, ease of cleaning), and the main competing solutions currently available (e.g., small steam distillers and multifunction kitchen appliances). In parallel, the study addressed the underlying extraction principles by reviewing the relevant physical models and the chemical–physical conditions required for essential oil recovery, including temperature and pressure ranges, phase-change phenomena, heat and mass transfer aspects, and the need for effective condensation and phase separation. Building on these inputs, the device architecture was defined and translated into a complete 3D CAD model developed in SOLIDWORKS, including the main assemblies and key functional components. The modelling phase was supported by iterative design choices oriented toward manufacturability, assembly/disassembly, maintenance, and safe operation in a household environment. A material selection activity was carried out to identify suitable options for each component, evaluating mechanical and thermal requirements, together with cost, availability, and compatibility with food-contact constraints where applicable. Finally, a preliminary environmental assessment was performed at component level, focusing on the impact associated with raw material extraction and manufacturing of the single parts, with the aim of highlighting the most impactful elements and outlining possible improvement strategies for a more sustainable design.

Why, How and What?

First, it is necessary to justify the choice of this product; the selection of a kitchen-scale essential oil extractor was motivated by the increasing use of essential oils in everyday applications, particularly in culinary preparations, personal care routines, and general household use (e.g., natural fragrances and cleaning solutions). This trend suggests a growing interest in small-batch, on-demand production and in greater control over the origin and quality of the extracted product.

A brief review of the current market also indicates that direct competitors are relatively limited, and the existing offer is only weakly differentiated, with most solutions relying on similar operating principles and comparable product architectures. Once the choice had been justified from a market-study point of view, we focused on the main physical and chemical principles that make the extraction possible, for this purpose we began by studying how a typical extractor works. Essential oil extraction in a traditional alembic is primarily based on steam distillation / hydrodistillation, which exploits the volatility of aromatic compounds and their typical immiscibility (or limited solubility) in water.

When water and essential oil are heated together, the system behaves as a co-distillation: the total vapour pressure is approximately the sum of the partial vapour pressures of water and the volatile organics (Dalton's law). As a consequence, the essential oil can reach the boiling condition at a temperature lower than the normal boiling point, limiting thermal degradation while still enabling their transfer to the vapour phase. In a traditional alembic, these principles are implemented through a sequence of functional sections:

1. Boiler : water is heated (with plant material either immersed—hydrodistillation—or supported above the water with steam passing through—steam distillation). Volatile compounds evaporate together with water vapour.
2. Vapour path / head: the vapour mixture flows upward, driven by buoyancy and pressure gradients.
3. Condenser (serpentine or water-cooled coil): the vapours are cooled below the dew point, undergoing phase change back to liquid.

4. Separator: the condensate splits into two layers due to density difference and limited mutual solubility: an essential oil phase and an aqueous phase (hydrosol). The oil is then collected by decantation/overflow, depending on whether it is lighter or heavier than water.

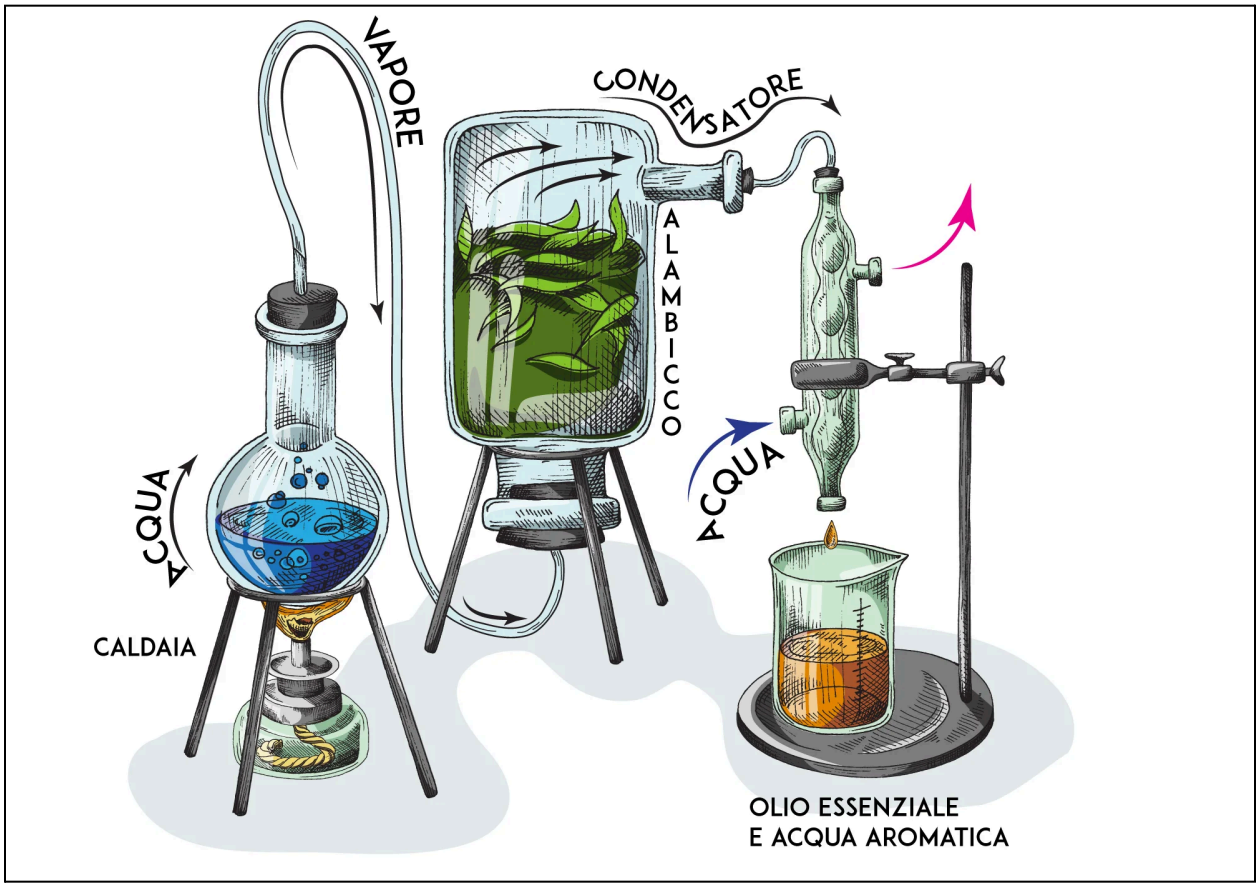


Figure 1. Basic Scheme of Oil Extraction

Once the need for a domestic essential-oil extractor had been established and the underlying extraction mechanism clarified, it was determined that the concept required a two-chamber architecture. The first chamber is dedicated to the heating and boiling stage, where the vapour mixture is generated, while the second chamber is designed for condensation, enabling the vapours to be cooled and converted back into liquid. The condensate is then collected in an integrated reservoir to allow safe handling and final withdrawal of the product.

Within this architecture, a dedicated thermal insulation system is also required. Insulation supports effective condensation by limiting unwanted heat losses and thermal cross-coupling between the hot and cold sections, improves overall thermal efficiency by confining heat where it is needed, and—crucially—prevents the external surfaces of the device from reaching temperatures that could be hazardous for the user.

These design choices were developed while maintaining appropriate safety standards, explicitly addressing the criticalities associated with elevated temperature and internal pressure, and while targeting a compact footprint suitable for kitchen use together with a renewed styling approach to differentiate the product from conventional solutions

3D CAD modelling

The CAD modelling phase proved to be one of the most challenging part of the project, mainly because the design task was approached as a variant design. In practice, no legacy technical drawings, reference CAD files, or validated geometries were available to be reused or adapted. As a result, the 3D model had to be developed from first principles, starting from functional requirements and preliminary layout assumptions. This lack of a predefined baseline led to several iterative redesign cycles during the modelling activity. As the assembly progressed and the level of detail increased, a number of physical and geometric constraints emerged—such as clearances, component interfaces, feasible wall thicknesses, routing of the vapour path, and packaging limitations—which required repeated adjustments to part geometries and assembly constraints in order to maintain consistency, manufacturability, and overall functional feasibility.

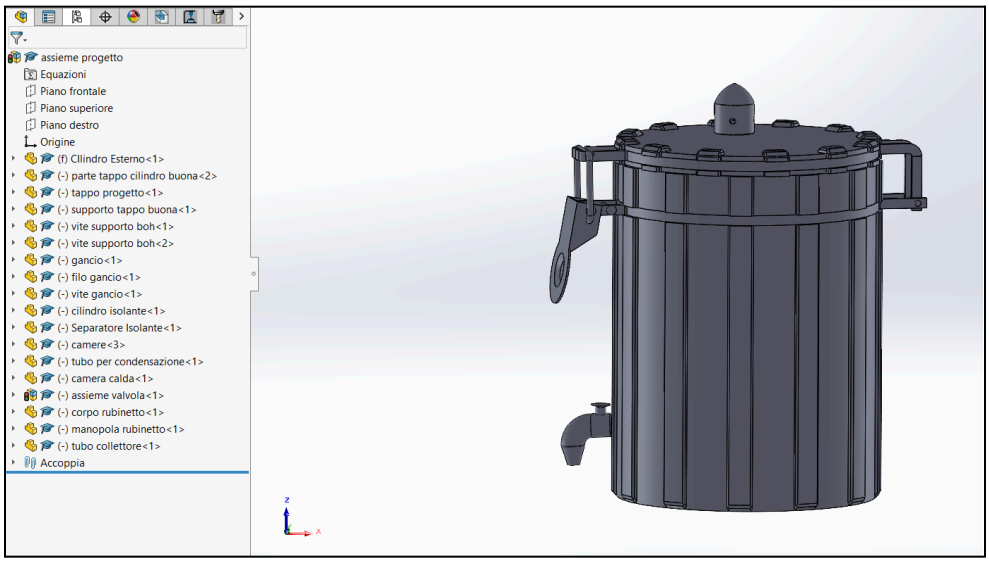


Figure 2. CAD Components & Assembly

The CAD model can be divided into multiple sub-assemblies, each designed around a specific function within the overall extraction workflow. For each sub-assembly, the design was defined by its role in the process (boiling, vapour transfer, condensation, collection, and safety management) and by its correspondence to elements typically found in a traditional alembic.

1. Thermal insulation system

This sub-assembly is aimed at limiting heat losses from the hot section and preventing the external surfaces from reaching unsafe temperatures during operation. From a process standpoint, it supports the boiling/extraction stage by improving thermal efficiency and, at system level, helps preserve the thermal gradient between the hot chamber and the cold chamber, which is beneficial for condensation. This sub-system is composed of an external cylinder which is interposed between the internal cylinder composed by the **hot** and **cold** chambers and the external core; and an internal wall between the two chambers.

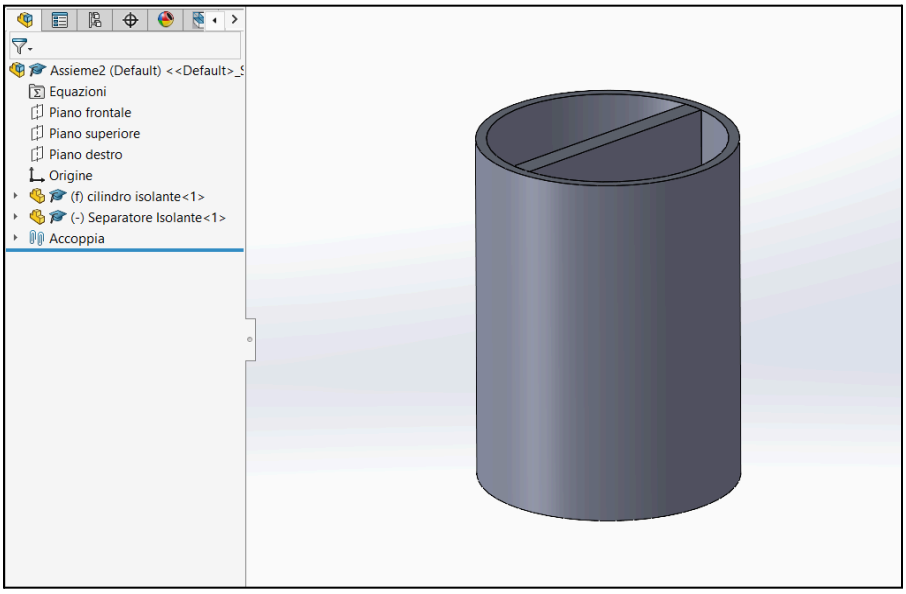


Figure 3. Insulation Components

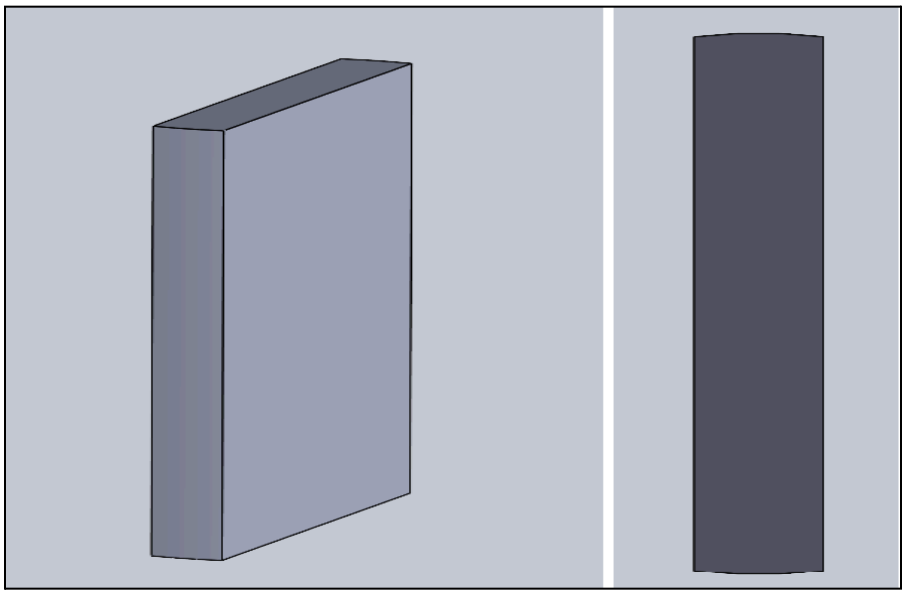


Figure 4. Insulation Components - internal wall between the two chambers

A supplementary document attached to the presentation includes the preliminary thermal analysis carried out for this sub-assembly, which has been subsequently validated through dedicated simulation tools such as ANSYS..

2. Safety system (pressure relief valve)

The safety sub-assembly manages the criticalities related to temperature and pressure by providing a controlled venting path. It consists of a vent/outlet and an internal spring-loaded mechanism, where the spring is sized according to the maximum allowable operating pressure of the device. Its role is to prevent overpressure conditions during the vapour generation phase, increasing user safety and protecting the structure.

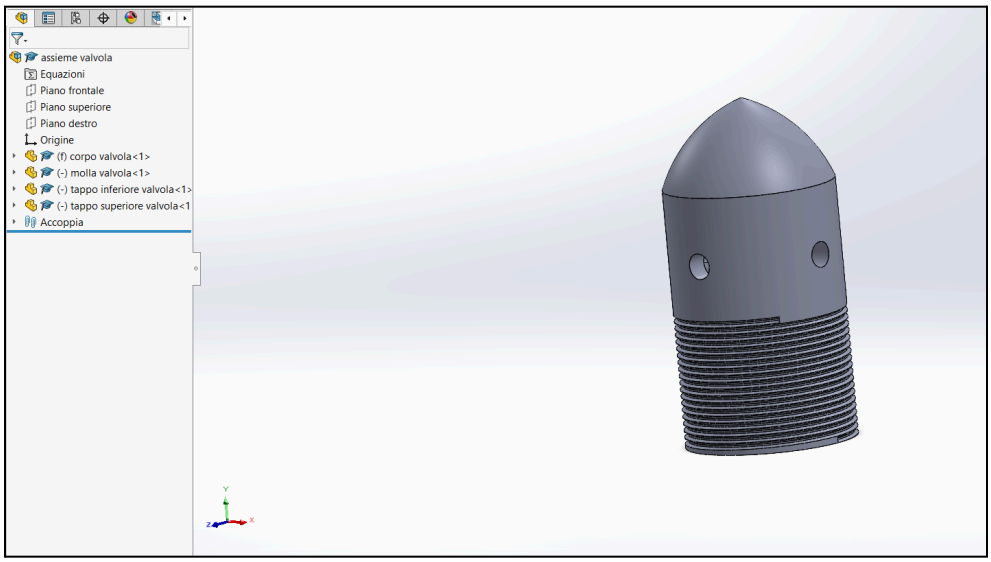


Figure 5. External View of Valve

In typical extraction conditions, the operating pressure does not exceed 2 bar, it means that the spring will have to open once this limit pressure is reached; considering this limit pressure the calculations to design this element are strictly related to an **equilibrium of forces** between internal pressure and the elastic force of the spring; taking into account a diameter of the plate on which the pressure is applied of $10\,mm$ we will have an applied force of :

$$F_p = 200.000 \times 7,85 \times 10^{-5} = 15,7\,N$$

we must also take into account a **safety factor of 1.5** that will lead to $Fp = 23,6\,N$ (in case the pressure reaches 3 bar); considering the Hooke's law, a preload and an opening displacement of 1mm a spring stiffness of approximately $k = 3,7\,\frac{N}{mm}$ **is required.**

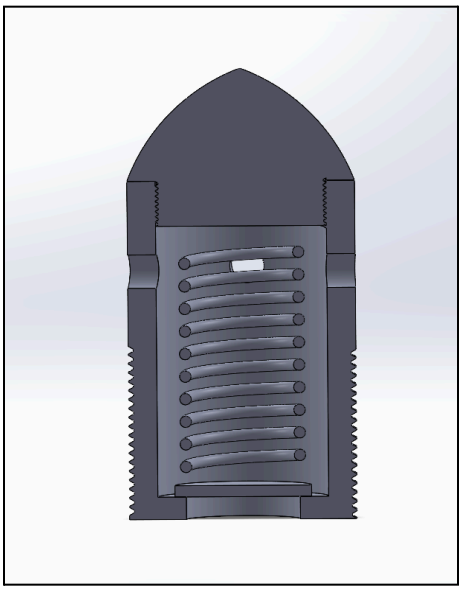


Figure 6. Valve Section

3. Support and connection system

This sub-assembly mechanically connects the lid to the central body, ensuring alignment, structural stability, and consistent sealing at the interface. It is directly related to the boiling and vapour containment stage, since it must withstand thermal loads and any internal pressure while enabling practical assembly/disassembly for cleaning and maintenance.

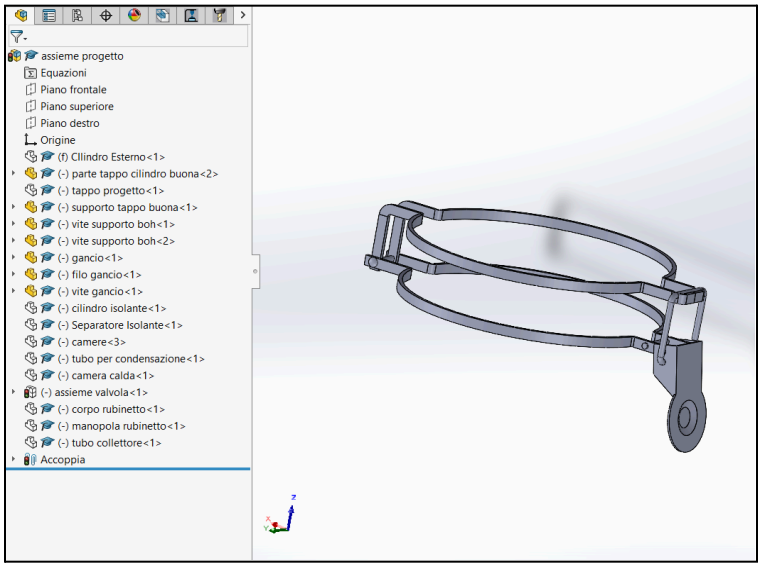


Figure 7. Cap Support Component

4. Transport and cooling system (vapour duct + condenser coil)

This sub-assembly is responsible for transferring the vapour mixture from the hot chamber to the cold chamber and for promoting heat exchange to achieve phase change. It consists of two ducts: the first conveys vapour from the boiling zone, while the second evolves into a serpentine coil inside the cold chamber, where condensation occurs and vapours return to the liquid state. It's very important to underline that in the transport and cooling sub-assembly, the phase change inside the condensation tube is ensured by the thermal conditions of the process itself. During the boiling stage, the vapour generated in the hot chamber does not exceed 111,4 °C (boiling temperature at 1.5 bar), since the supplied energy is predominantly absorbed as latent heat to sustain evaporation rather than increasing the vapour temperature. Consequently, even a relatively small temperature gradient between the vapour stream and the cold chamber (and, by conduction, the tube wall) is sufficient to cool the vapour below its saturation temperature. Once the vapour temperature drops below approximately 111,4 °C, condensation occurs along the serpentine, enabling the transition back to the liquid phase and subsequent collection.

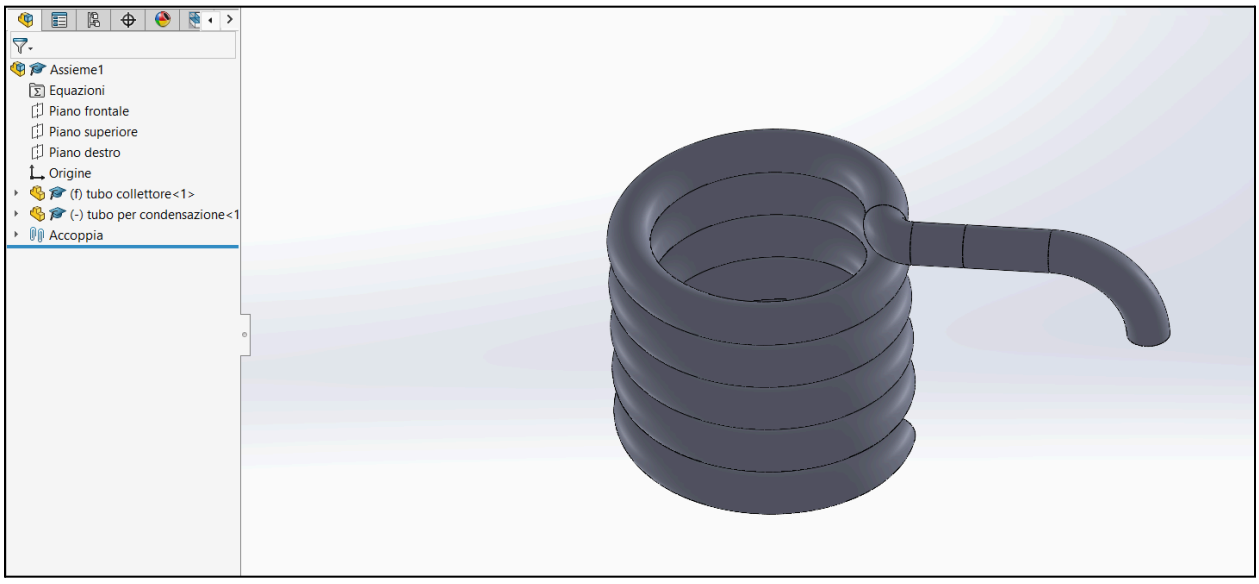


Figure 8. Vapor Duct + Condenser coil

5. Operational core: “hot” chamber and “cold” chamber

This is the main functional sub-assembly of the device. In the hot chamber, vapour generation is achieved by means of an immersion heating element, calibrated to provide the thermal power required to bring water to boiling and sustain a stable evaporation rate. The selected extraction method is steam distillation, as it allows essential oils to evaporate at lower effective temperatures, thereby preserving delicate aromatic compounds and reducing the risk of thermal degradation, at this aim has been mounted a perforated grill on which the product will be placed and steam will pass through the holes performing the extraction. Afterwards, the cold chamber hosts the condensation environment and integrates the collection features: the condensed liquid is guided by an inclined surface towards an outlet tube connected to a tap, allowing controlled withdrawal of the distillate for subsequent handling and separation. The tap will be made, at least partly, of glass so that the user can distinguish between oil and hydrosol.

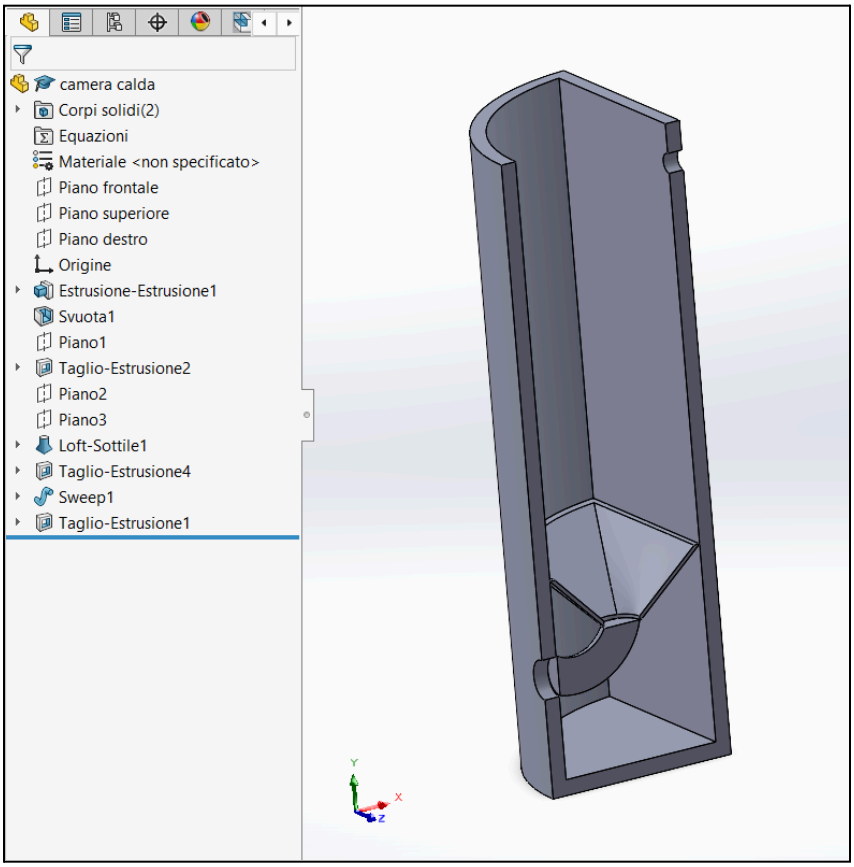


Figure 9. Cold Chamber

It is necessary to specify that both chambers and their components (grid, collector, and duct) are made of **AISI 304**.

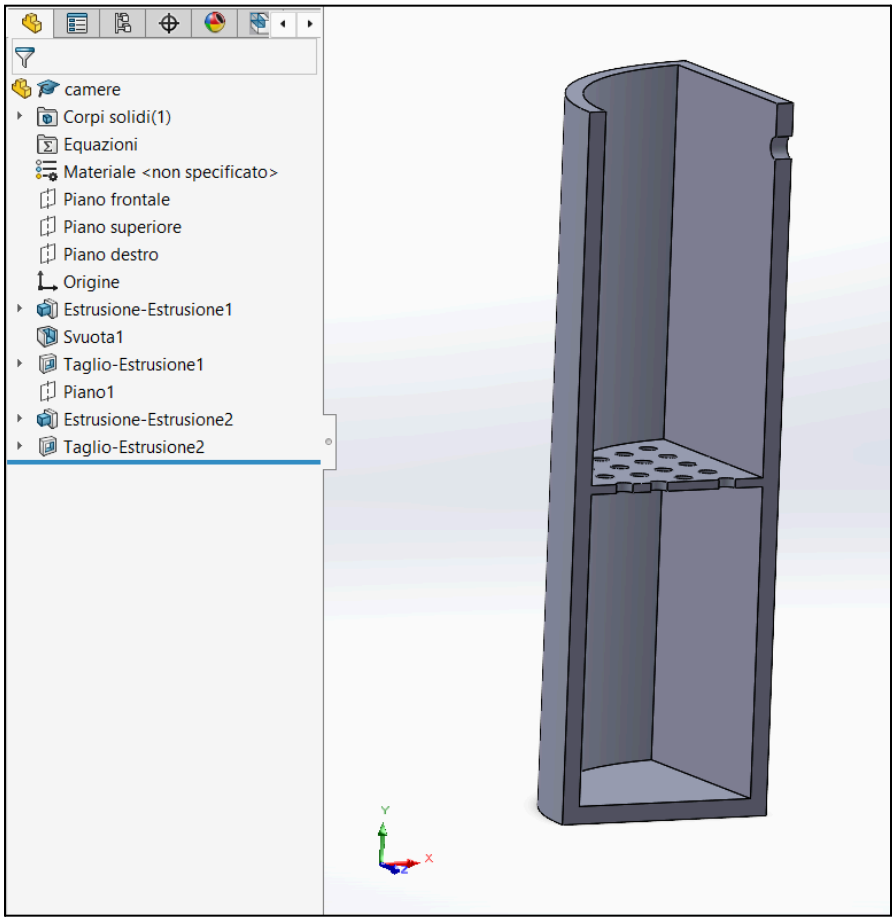


Figure 10. Hot Chamber

The required power of the heating element was estimated by considering a volume of water equal to the volume of the portion of the cylinder below the grid. Assuming a boiling time greater than or equal to seven minutes, the resulting power requirement is approximately 1.2 kW, which is easily achievable with commercially available components and straightforward to integrate into the extractor. It is important to note that the heating element and the associated electrical components are not represented in this model due to their secondary importance and geometric complexity.

6. Outer core

An additional sub-assembly is represented by the external core of the container and the tap. Their primary function is to maintain the lowest possible external surface temperature, both to improve thermal efficiency (by reducing heat dissipation to the environment) and to increase user comfort and safety during handling.

In the early design stages, those components were assumed to be made of AISI 304, given its widespread adoption in culinary and food-related products. However, due to its relatively high thermal conductivity and high density, the outer housing alone would have resulted in a mass close to 2 kg, with the additional drawback of facilitating heat transfer to the external surface. For these reasons, the design was revised in favour of **polymeric** materials, which are less aesthetically "premium" but more practical for this application. In particular, **polypropylene** was selected, as it is commonly used in domestic appliances such as electric kettles thanks to its low density, favourable thermal insulation behaviour, and suitability for consumer-grade manufacturing.

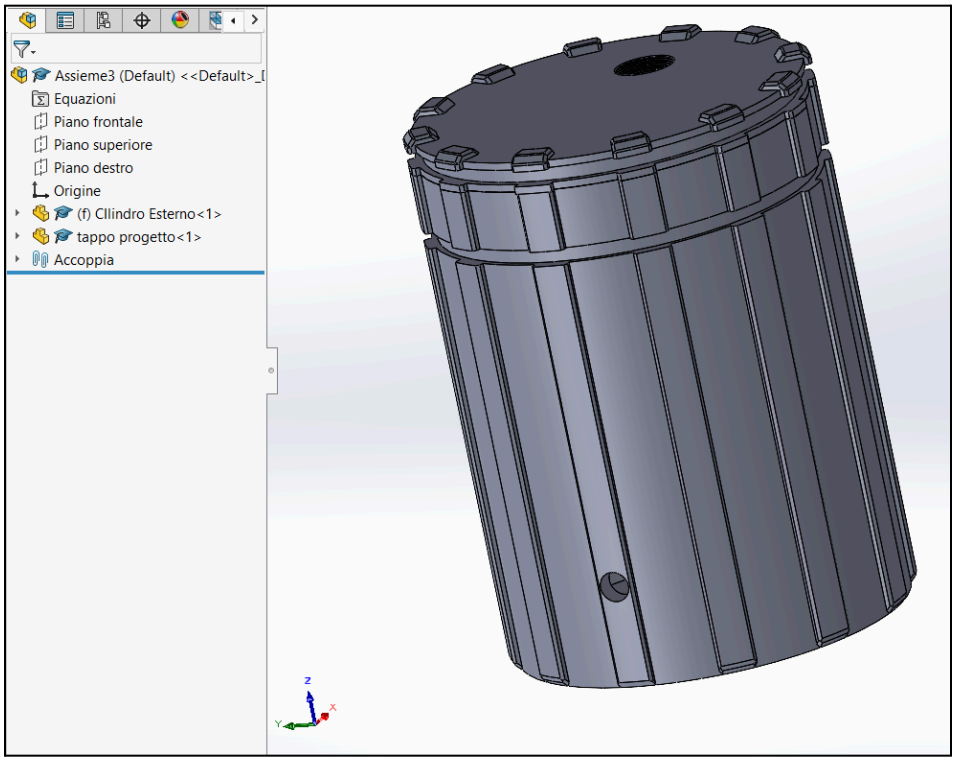


Figure 11. Outer core

In addition, the external “protrusions” serve both aesthetic and technical purposes: they increase the external surface area and, consequently, the heat-exchange surface, allowing for slightly more efficient cooling.

Validation

This chapter focuses on the thermal validation of the proposed design, with the aim of demonstrating that the selected materials, dimensions, and overall geometry satisfy the requirements defined at the beginning of the project in terms of user safety and thermal efficiency. During the design process, we recognised that the thermal behaviour of the system could be reliably characterised using standard heat-transfer tools from classical thermal engineering. The main geometrical features can in fact be reduced to simple domains (cylindrical walls and flat partitions), for which heat-transfer correlations and 1D thermal-resistance models provide sufficiently accurate predictions for early-stage design.

For this reason, the thermal sizing and verification were primarily performed through a dedicated MATLAB script. To strengthen the robustness of the results, a set of independent numerical simulations was subsequently carried out using ANSYS, confirming the validity of the analytical estimates and the consistency of the predicted temperature distributions. This two-step approach allowed us to combine the efficiency of analytical modelling with the additional confidence provided by high-fidelity simulation.

The full set of outputs is reported in the attached technical note; the key results are summarised below.

1. External safety: a 5 mm **polyurethane (PU)** insulation layer limits the external wall temperature to an average of 41 °C, satisfying the user-safety requirement. An additional 5 mm **polypropylene** layer further reduces the overall weight and enhances the thermal insulation, contributing to lower external surface temperatures and improved handling safety.
2. Thermal decoupling between chambers: a **10 mm PU** insulation layer between the two chambers provides an almost complete thermal separation, with an estimated heat transfer of about 1 W across the partition.

3. Overall efficiency: total heat losses (curved external walls plus internal partition) remain below 10 W, indicating a thermally efficient system with limited energy dispersion.

From a structural standpoint, no critical issues are expected. The most loaded region corresponds to the two internal chambers, where the maximum pressure is limited to 2 bar by the safety valve. This pressure level is significantly below the yielding threshold of AISI 304; at 111.4 °C, the material yield strength is approximately 155 MPa, providing a wide safety margin for the operating conditions considered.

In light of these considerations, and following the same cost-effectiveness principle adopted for the thermal assessment, a dedicated FEA-based structural simulation would not be strictly necessary at this stage. Nevertheless, an ANSYS verification was performed for completeness and to further support the robustness of the design choices. The analysis focused on the most critical loading condition, combining (i) the maximum internal overpressure of 2 bar—corresponding to the opening set-point of the safety valve—and (ii) the thermal load associated with the hot chamber, modelled as the heat flux generated by the vapour during operation.

To ensure feasibility within the constraints of the ANSYS Student license, the CAD geometry was simplified: all internal components within the two chambers were removed, and the external cylinder was stripped of purely aesthetic features. This approach allowed the analysis to concentrate on the load-bearing pressure boundary while avoiding meshing issues associated with small details that would have required an excessively fine discretisation.

The simplified geometry was imported into ANSYS Workbench and the main materials were defined accordingly. In addition to AISI 304 (Poisson's ratio 0.30, Young's modulus 190 GPa in the relevant temperature range), polyurethane and polypropylene were included with their respective thermo-mechanical properties. This enabled the model to capture not only the metallic pressure boundary but also the contribution of the insulating layer and the external housing in terms of thermal behaviour and component interaction.

A mesh was generated to balance accuracy with the available computational resources, after which boundary conditions and loads were applied (constraints on the relevant

interfaces and internal pressure on the wetted surfaces). The resulting stress distributions show that the highest stress levels occur at internal sharp edges and, locally, in the central region of the wall adjacent to the insulating partition. The peak values at the edges are consistent with numerical stress concentrations due to idealised sharp corners; in the real component these features are filleted, which would reduce the maximum stresses. In all cases, the computed stresses remain well below the yielding threshold, confirming a wide safety margin and an overall satisfactory structural behaviour.

From a thermal standpoint, the ANSYS results are fully consistent with the MATLAB-based calculations. They confirm effective insulation both between the two chambers and between the chambers and the external cylinder, resulting in limited heat transfer across the separation wall and low external surface temperatures. This outcome validates the insulation sizing and supports the user-safety requirements during normal operation.

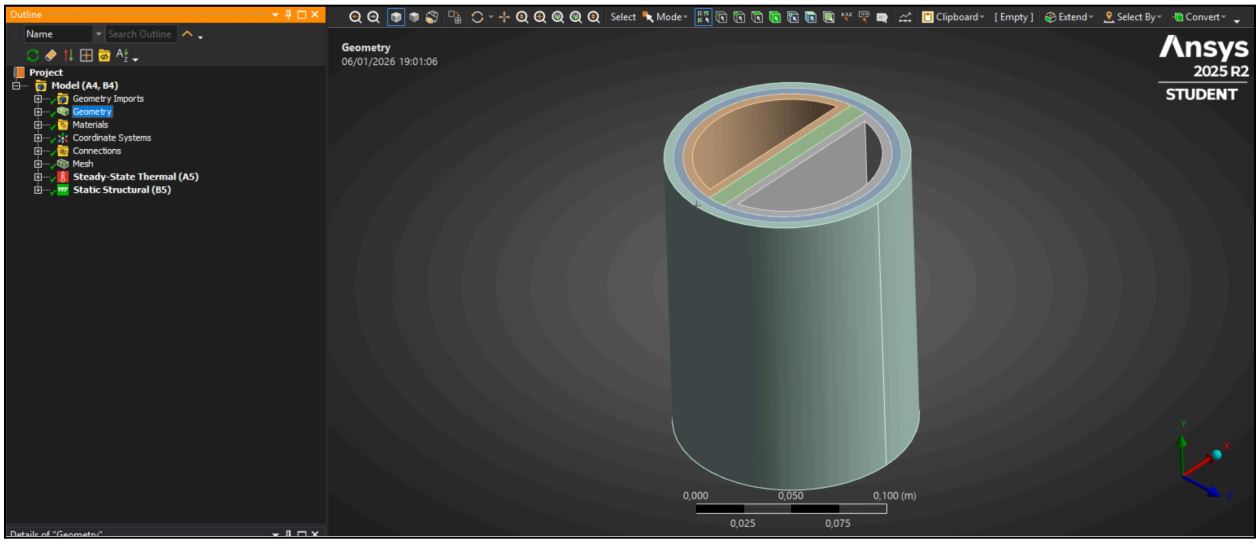


Figure 12. Simplified Geometry for Ansys

Material	Thermal conductivity $[W⍰(mK)^{-1}]$
AISI 304	$0,16$
Polyurethane foam	0.022
Polypropylene	$0,017$

Tab 1. Materials thermal property for Ansys Thermal analysis

Material	Young Modulus $[GPa]$	Poisson ratio [\]
AISI 304	190	$0,30$
Polyurethane foam	$0,02$	$0,30$
Polypropylene	$1,2$	$0,42$

Tab 2. Materials mechanical properties for Ansys Mechanical analysis

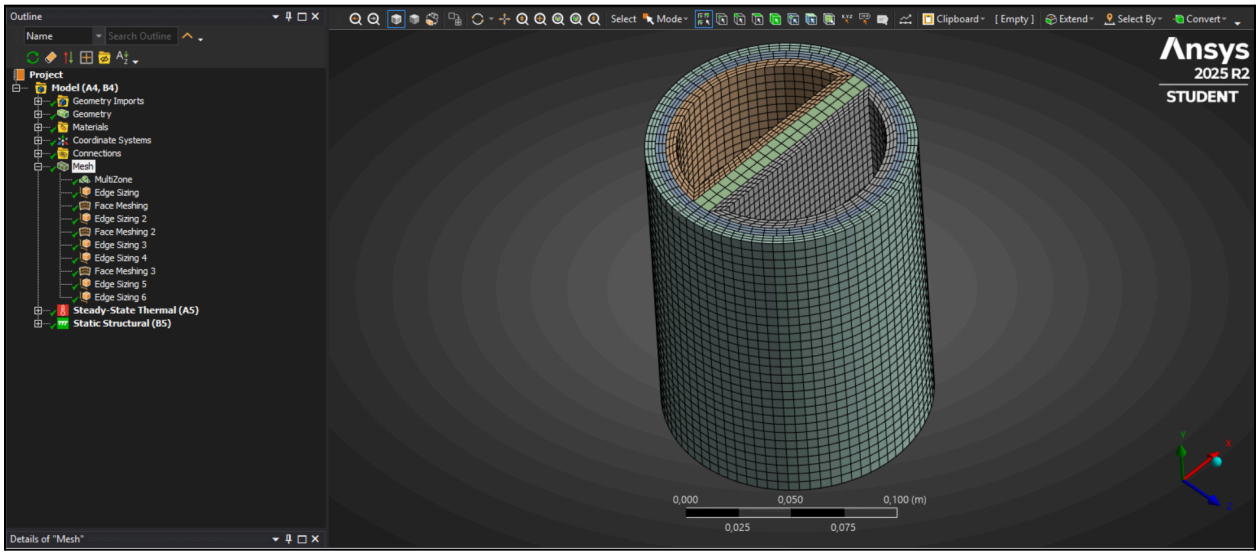


Figure 13. Meshed Component

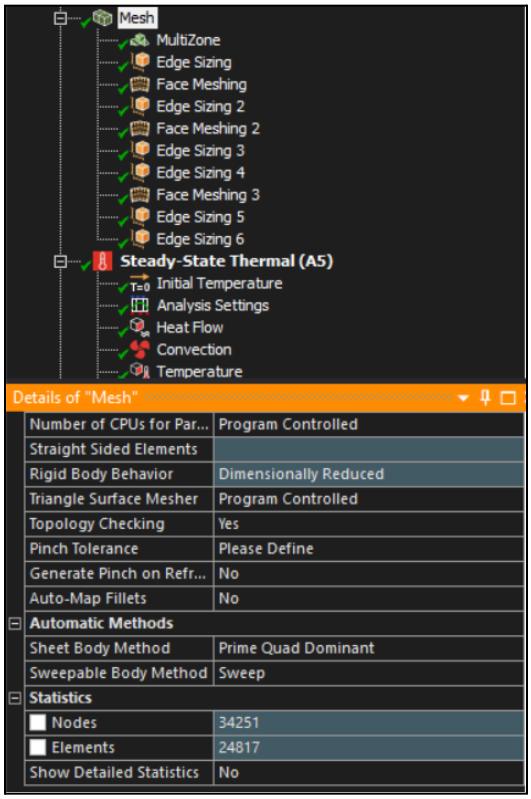


Figure 14. overall mesh informations

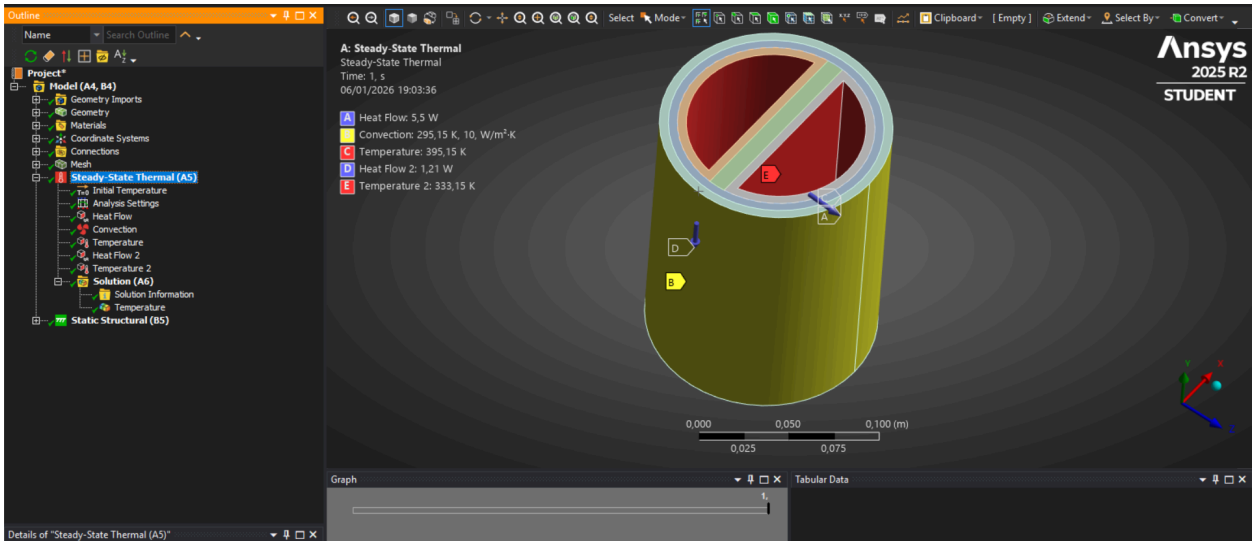


Figure 15. Thermal boundary conditions

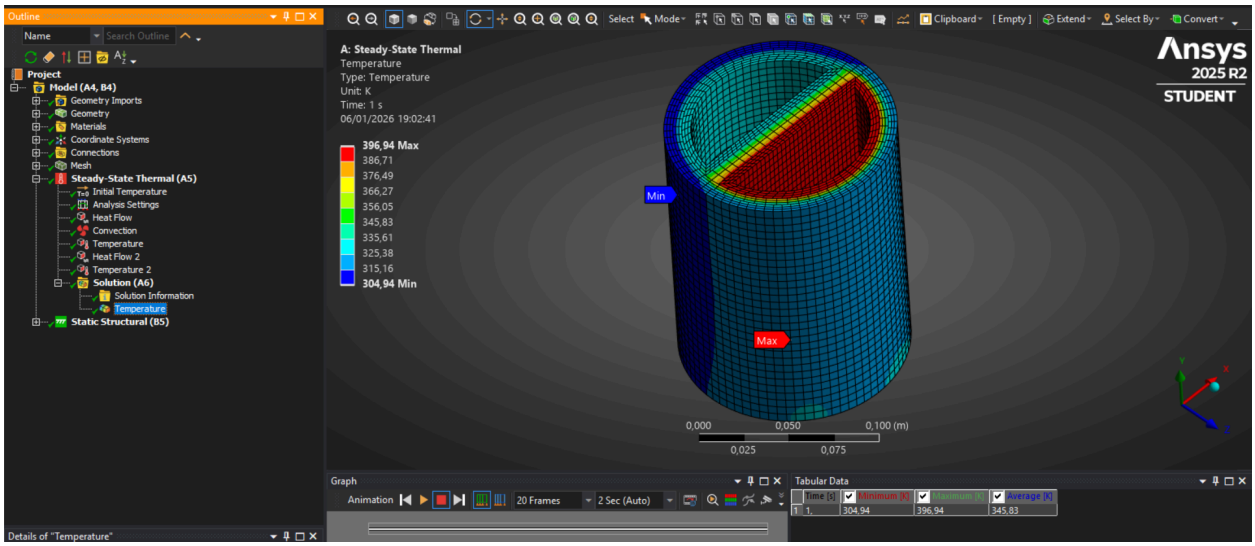


Figure 16. Temperature's trend

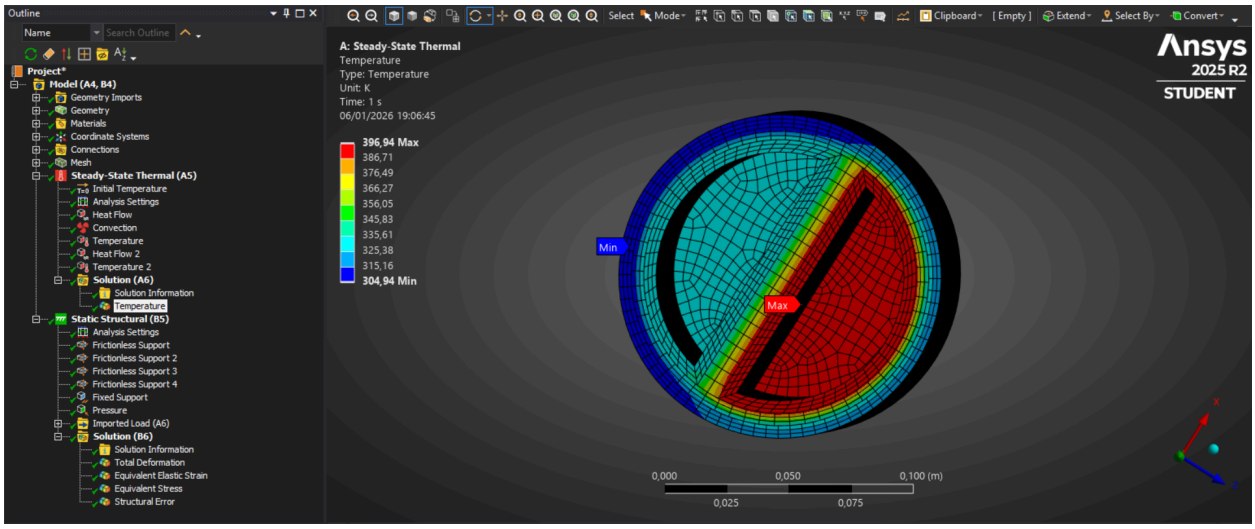


Figure 17. Maximum and minimum temperatures

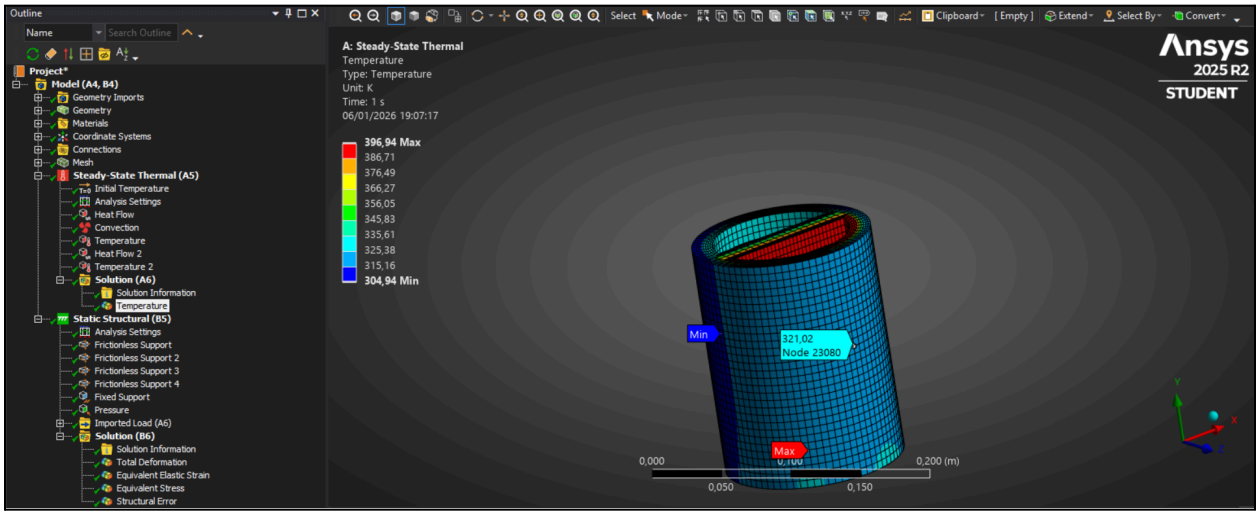


Figure 18. maximum temperature at the outer surface

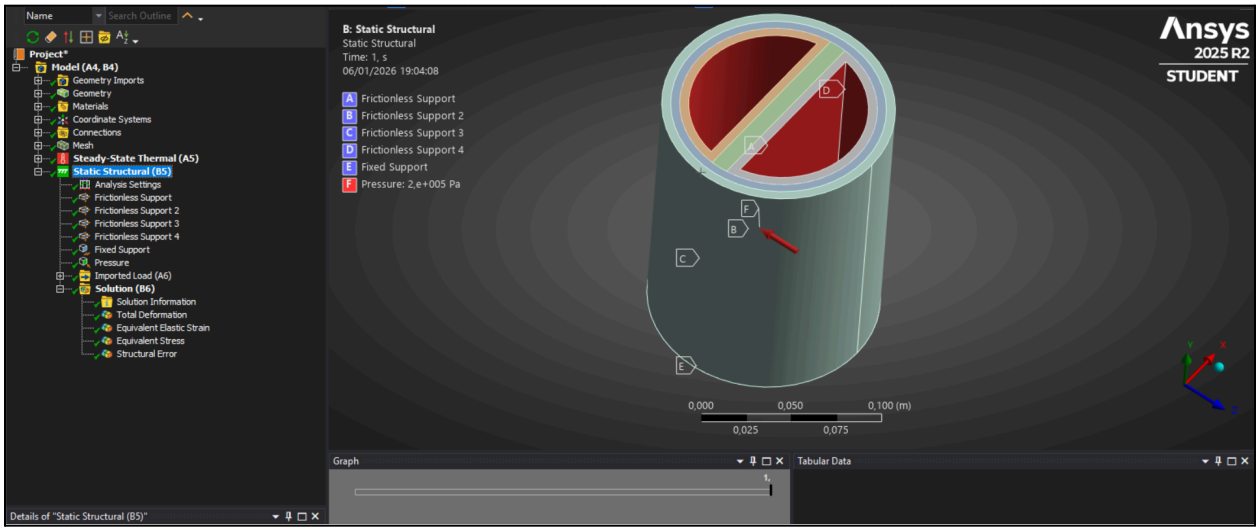


Figure 19. Mechanical boundary conditions

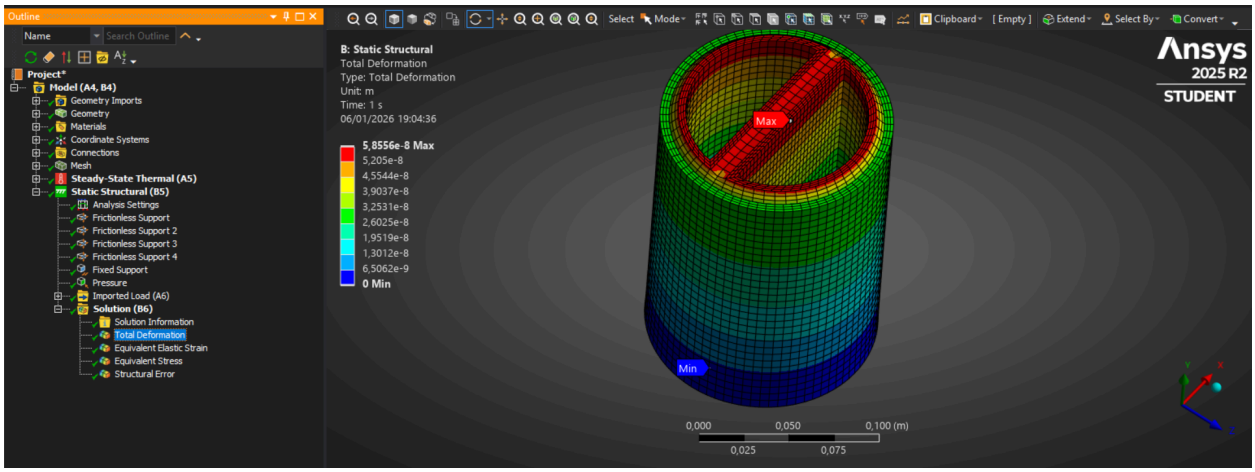


Figure 20. Total strain's trend

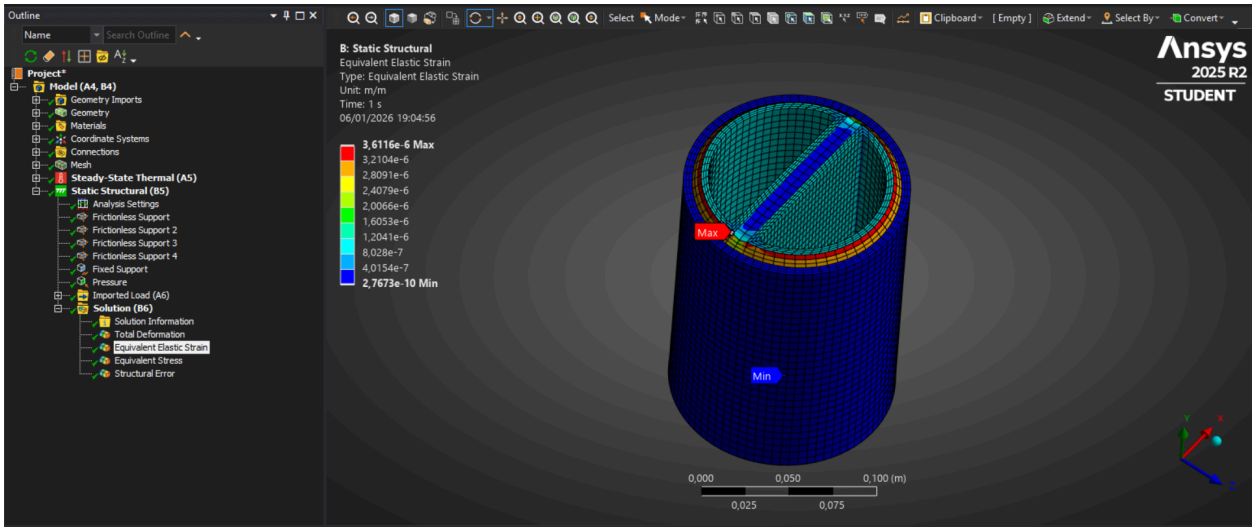


Figure 21. Strain's trend

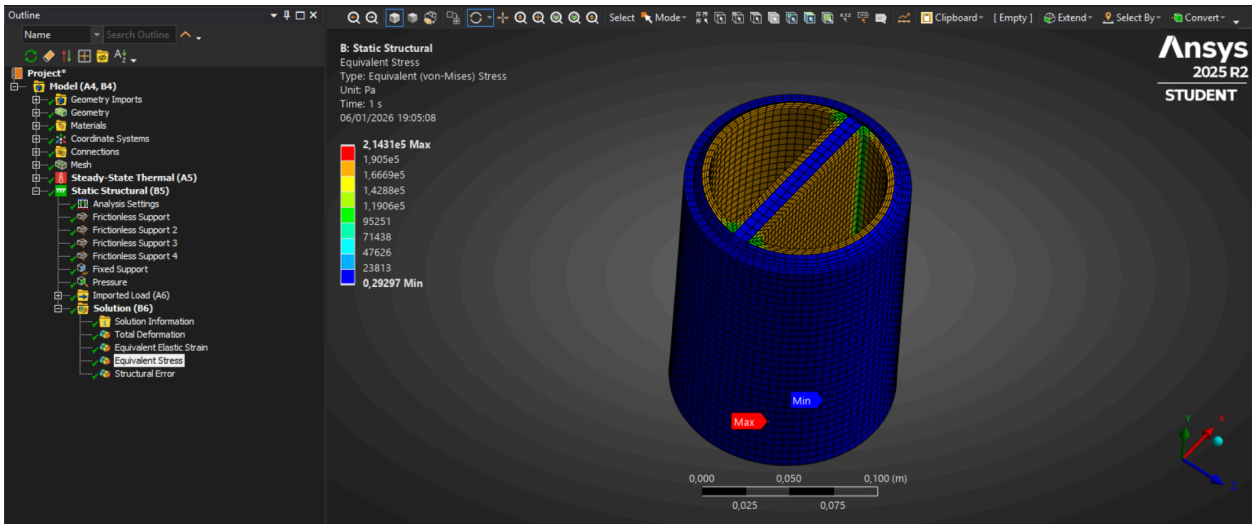


Figure 22. Stresses trend

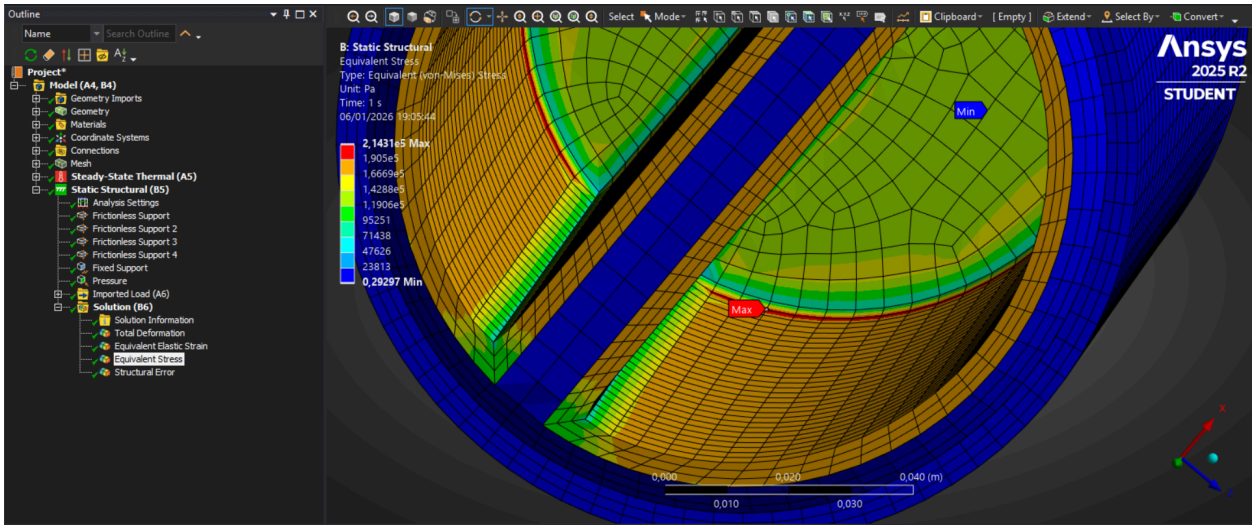


Figure 23. Maximum stress

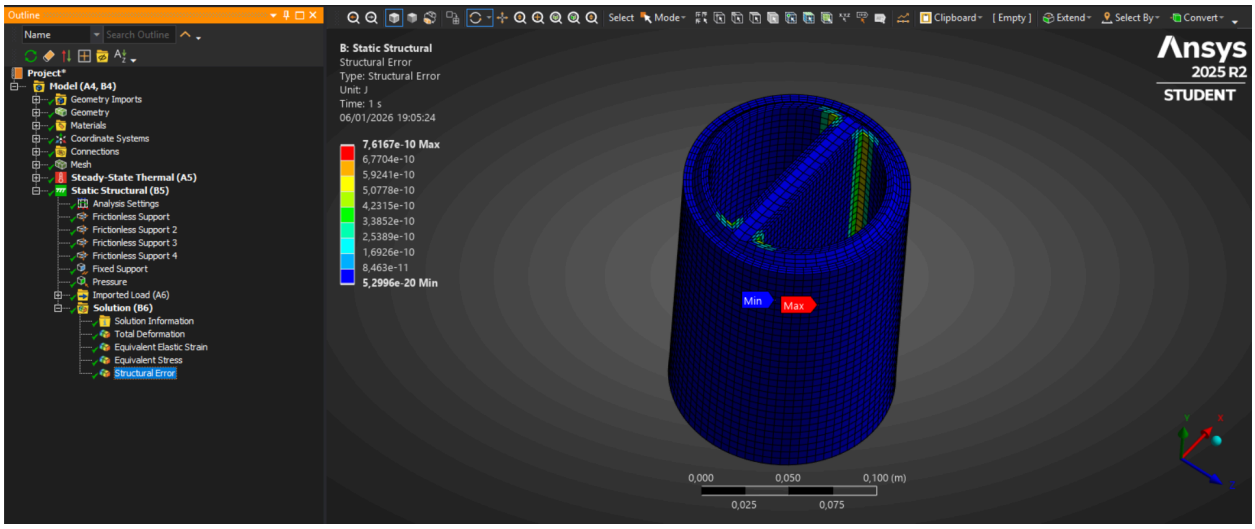


Figure 24. Safety factor trend

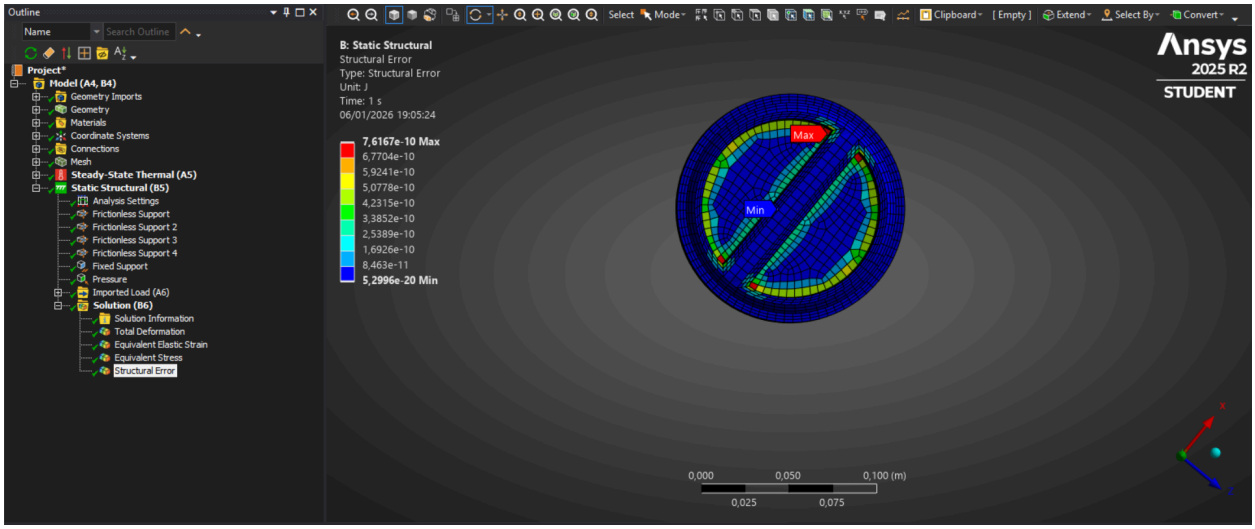


Figure 25. Lowest safety factor's trend

	$T\ [K]$	*T [°C]*
Maximum temperature (Hot chamber)	$396,94$	$123,79$
Average temperature Outer Shell (Hot Side)	$315,16$	*42*
Maximum temperature (Cold Chamber)	$325,38$	$52,23$

Tab 3. Resume of results of Therma Ansys analysis

	$\sigma[Pa]$
Maximum stress	$2,1431 \cdot 10^{5}$
Minimum stress	$0,29297$

Tab 4. Resume for stresses results of Mechanical Ansys analysis

	ε *[\]*
Maximum strain	$5,8586 \cdot 10^{-8}$
Minimum strain	0

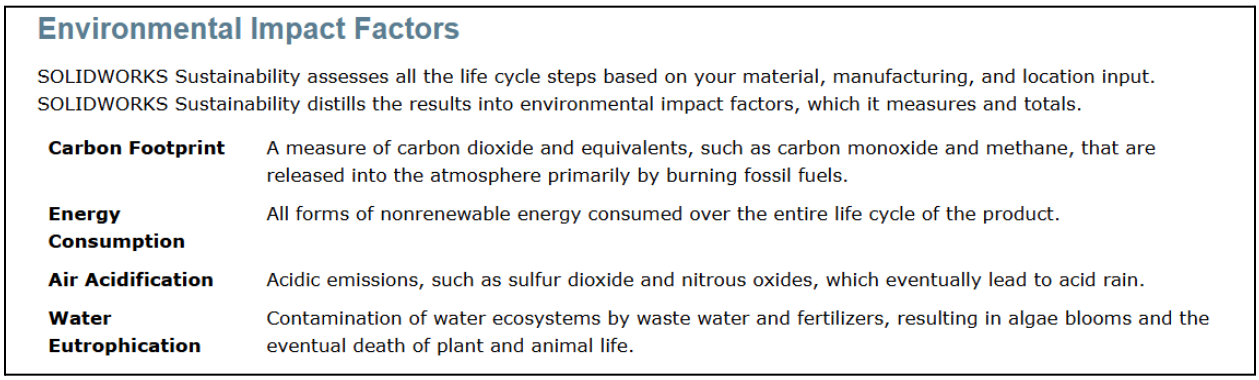
Tab 5. Resume for strains results of Mechanical Ansys analysis

	S.F [\]
Maximum S.F	$7,6167 \cdot 10^{-10}$
Minimum S.F	$5,2996 \cdot 10^{-20}$

Tab 6. Resume for safety factors results of Mechanical Ansys analysis

Preliminary LCA

Before the final conclusions, an additional chapter was dedicated to a preliminary assessment of the product's environmental footprint and manufacturing cost using the SOLIDWORKS Sustainability tool.



Environmental Impact Factors

SOLIDWORKS Sustainability assesses all the life cycle steps based on your material, manufacturing, and location input. SOLIDWORKS Sustainability distills the results into environmental impact factors, which it measures and totals.

Carbon Footprint	A measure of carbon dioxide and equivalents, such as carbon monoxide and methane, that are released into the atmosphere primarily by burning fossil fuels.
Energy Consumption	All forms of nonrenewable energy consumed over the entire life cycle of the product.
Air Acidification	Acidic emissions, such as sulfur dioxide and nitrous oxides, which eventually lead to acid rain.
Water Eutrophication	Contamination of water ecosystems by waste water and fertilizers, resulting in algae blooms and the eventual death of plant and animal life.

Figure 26. Solidworks Sustainability Analysis

The analysis was performed with a cradle-to-gate approach, from raw material production to the finished component, and it accounts for transportation, potential process losses, end-of-life scenarios (including material recyclability), and possible disposal costs. The electricity and fuel consumption data associated with the production of each selected material (polyurethane, polypropylene, and AISI 304) were taken from the reference documents reported in the Annexes. The SOLIDWORKS Sustainability output reports four environmental impact indicators (carbon footprint, energy consumption, air acidification, and water eutrophication) and breaks them down by life-cycle stage (material, manufacturing, transport, use, and end-of-life). With the current assumptions (3-year use phase, with no operational energy modelled), the use-phase contribution is null in all categories, and the impact is dominated by upstream stages.

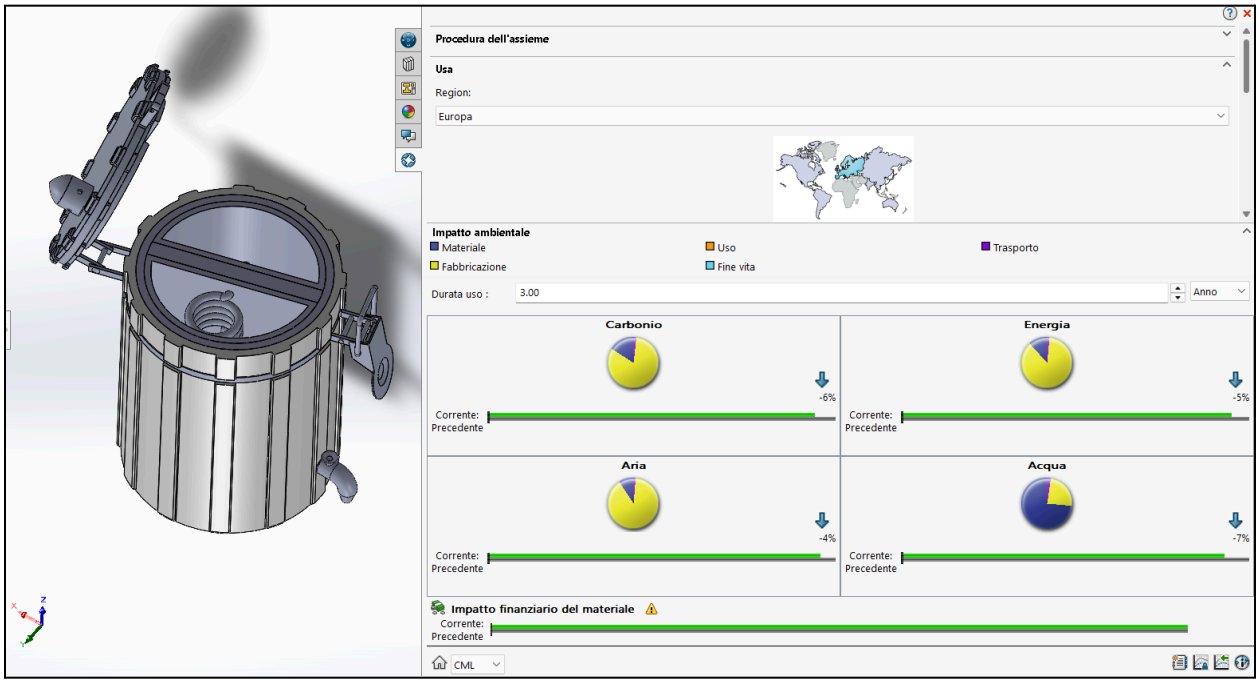


Figure 27. Overall environmental footprint

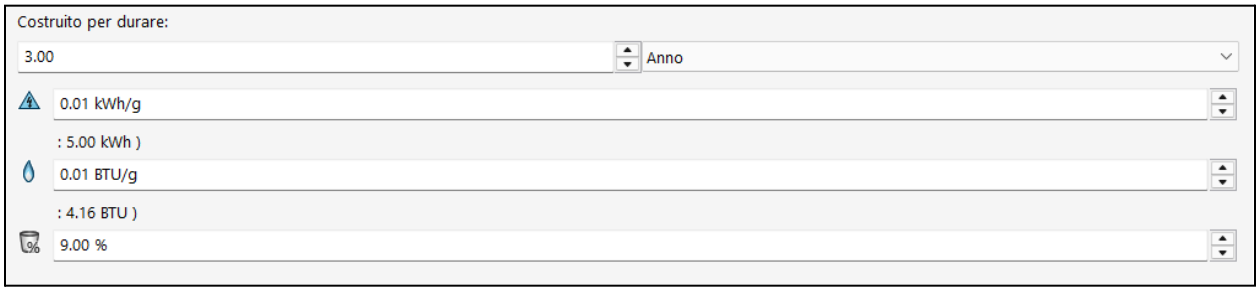


Figure 28. Energy consumption interface

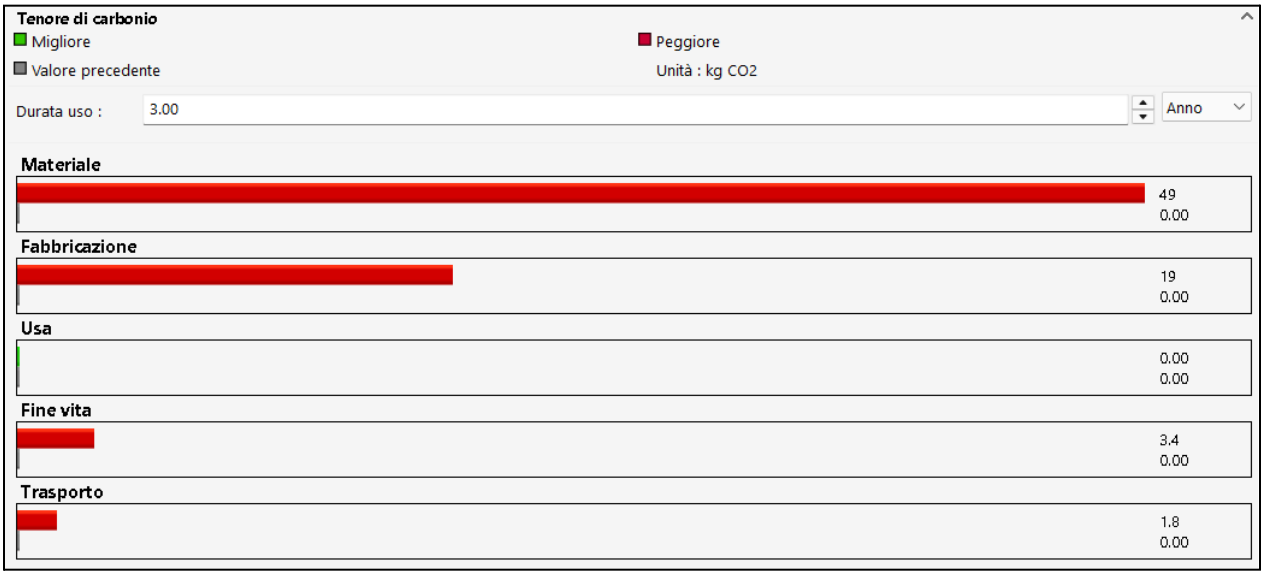


Figure 29. Carbon footprint (kg CO2)

Contribution source	kg CO2
Materials	49
Manufacturing	19
Transport	1.8
End of life	3.4
Total	73.2

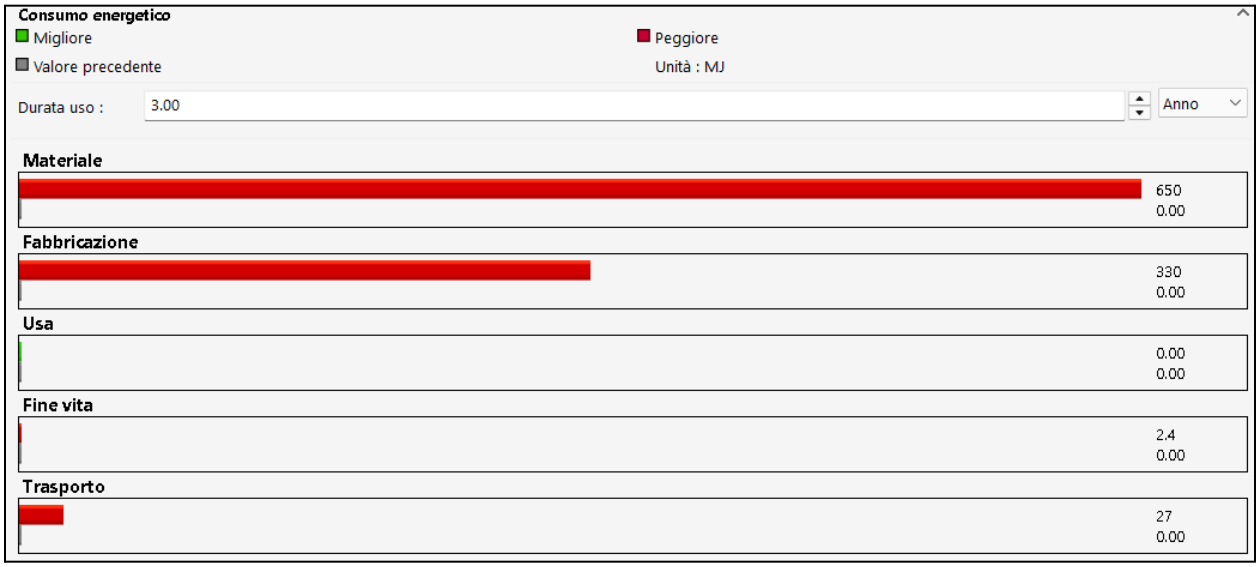


Figure 30. Energy consumption (MJ)

Contribution source	Energy consumption (MJ)
Material	650
Manufacturing	330
Transport	27
End of life	2.4
Total	1009.4

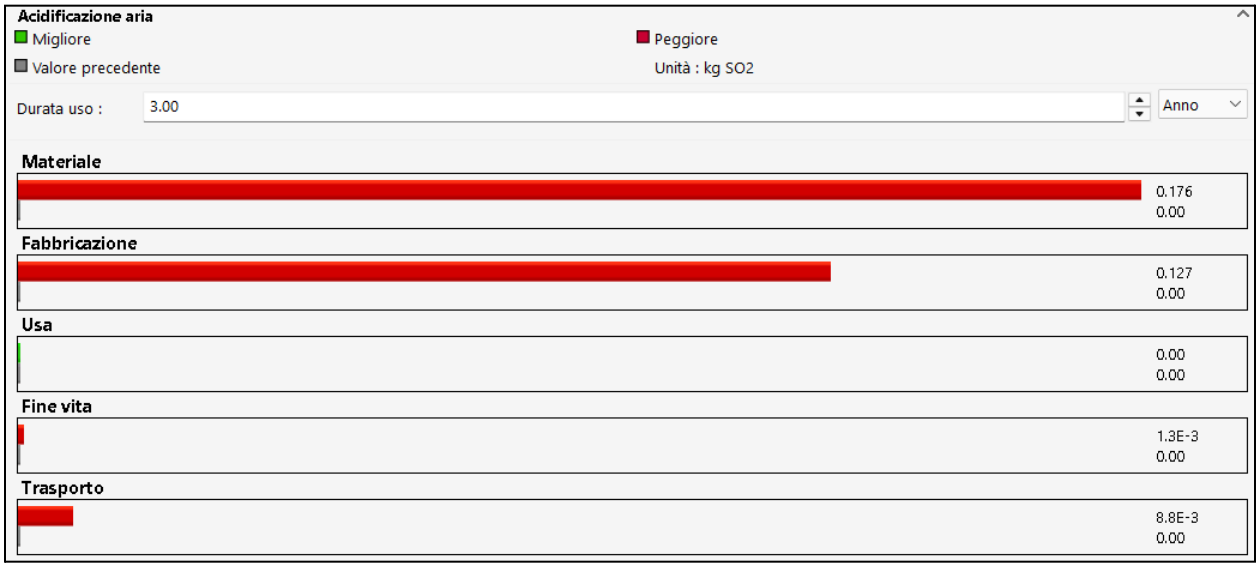


Figure 31. Air acidification (kg SO2)

Contribution source	Air Acidification (kg SO2)
Material	0.176
Manufacturing	0.127
Transport	8.8e-3
End of Life	1.3e-3
Total	0.313

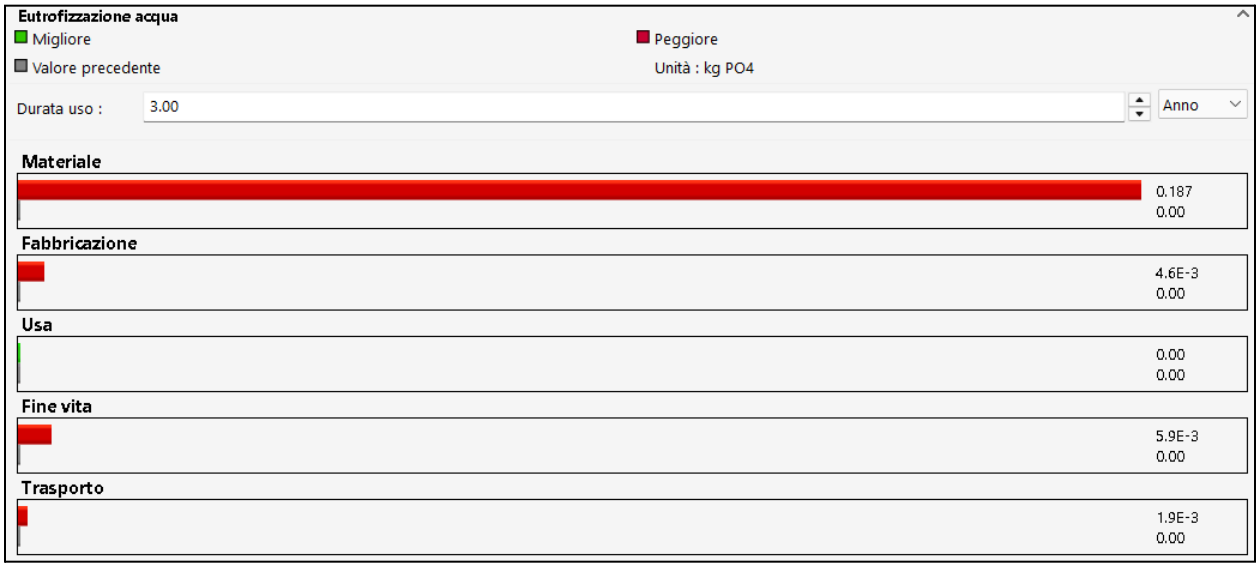


Figure 32. Water eutrophication (kg PO4)

Contribution source	Water eutrophication (kg PO4)
Material	0.187
Manufacturing	4.6e-3
Transport	1.9e-3
End of Life	5.9e-3
Total	0.199

To conclude, these results confirm that the main environmental burden is primarily associated with raw material extraction and production, followed by the manufacturing processes required to obtain the final component geometries. Transport and end-of-life contributions remain comparatively limited under the selected assumptions.

Conclusion

In conclusion, this project covered the main phases of a structured product development process for a kitchen-scale essential oil extractor. Starting from a brief market analysis, we identified a growing interest in essential oils for culinary, personal care, and domestic applications, and we highlighted a limited and weakly differentiated competitive landscape. Based on these considerations, we defined the functional requirements and studied the physical and chemical principles of steam distillation, translating them into a compact two-chamber architecture where vapour generation occurs in the hot chamber and condensation and collection take place in the cold chamber.

The concept was then implemented through a complete CAD development in SOLIDWORKS, organised into functional sub-assemblies (thermal insulation, safety valve, structural supports, vapour transport and condensation system, chambers and collection system, and external housing). In parallel, material selection was carried out to balance structural performance, cost, manufacturability, and user interaction, leading to the adoption of AISI 304 for the pressure boundary and polymeric solutions (polyurethane and polypropylene) to improve insulation, handling comfort, and overall practicality.

The design was validated from a thermal perspective using MATLAB-based calculations supported by an additional ANSYS verification: the results confirmed adequate external surface temperatures (around 41 °C), strong thermal decoupling between chambers (approximately 1 W across the partition), and limited overall heat losses (below 10 W), together with a structurally safe behaviour under the maximum operating pressure of 2 bar. Finally, a preliminary sustainability and cost assessment was carried out using the SOLIDWORKS Sustainability tool. The results indicate that the environmental impact is mainly driven by raw material production and by the manufacturing processes required to obtain the final component shapes, while transport and end-of-life contributions are comparatively smaller under the selected assumptions. The same analysis also provided an indicative manufacturing cost of approximately 24 EUR per component. Overall, in our opinion, the combination of a justified market need, innovative design choices, ease and comfort of use, and a strong emphasis on safety and sustainability makes the proposed device an example of product design and virtual prototyping.

Annex

1. https://www.acciaiterni.it/wp-content/uploads/2018/02/304.pdf
2. https://www.researchgate.net/figure/Effect-of-electricity-source-on-total-energy-for-stainless-steel-production_fig4_242689888
3. Consumi di energia e impatto ambientale della produzione dei ...
4. 1D Thermal Analysis